

Siddhant Bhardwaj

New Delhi, India | siddhantbhardwaj47@gmail.com | +91 87663 82920 | LinkedIn | GitHub | LeetCode

Summary

B.Tech CSE '27 backend engineer who has independently designed and shipped production systems – concurrency-safe transaction logic, distributed synchronization with exactly-once delivery, and APIs measured under load – as the sole backend owner at an early-stage payments startup. Comfortable owning a service from schema design through deployment and monitoring.

Education

Thapar Institute of Engineering and Technology

Sept 2023 – June 2027

- **B.Tech, Computer Science and Engineering** | CGPA: 8.35/10 | Finalist, BIS Hackathon
- **Coursework:** Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, OOP
- **Problem Solving:** 340+ algorithmic problems solved in C++ on LeetCode

Experience

Backend Developer – CosmicAttire

April 2026 – August 2026

Sole backend engineer at an early-stage startup building hardware-integrated identity verification and contactless payments

- Owned backend architecture end-to-end as the only backend engineer, building an IoT-to-server synchronization service that polled distributed ESP32 readers and processed buffered NFC tap events with **idempotency guarantees for exactly-once delivery** during network partitions.
- Designed a three-layer offline-first verification pipeline (server validation, LRU edge cache, batch-sync recovery) and a decoupled event-driven notification microservice on Supabase Edge Functions and Database Webhooks, isolating side-effects from the core Express API.
- Implemented **AES-256-GCM** payload encryption with MAC-based device authentication, and modelled indexed PostgreSQL schemas for hardware telemetry and sync state.

Product Development Intern – Goodmeetings.ai

July 2025 – August 2025

- Built and shipped a voice-driven onboarding bot (STT → LLM → TTS pipeline) enabling customers to configure and query dashboards in natural language, tuning pipeline stages to hold production latency requirements; worked directly with product managers to scope open-ended requirements into a shipped feature.

Projects

HostelBite *Multi-role campus food-ordering platform*

cemlus/hostel-bite | Live

Tools Used: Node.js, Express.js, MongoDB, JWT, Helmet.js, k6, OpenRouter (Gemini)

- Diagnosed a MongoDB query bottleneck causing 13.08s peak latency and a 24.98% failure rate under load, and eliminated it with compound indexing – cutting peak latency **80%** (to 2.62s) and raising throughput **27%**, validated via k6 across 4,900+ requests at a 0.00% error rate.
- Engineered atomic, sequential inventory decrement logic (MongoDB \$inc) with manual rollback on failed transactions; verified via k6 stress testing across 30 concurrent checkouts with **zero race conditions** and zero overselling.
- Implemented JWT authentication and role-based access control across Student, Shop Owner and Admin roles, hardened the API with security headers, CORS and rate limiting, and integrated GenAI for product listings.

GridNode *Decentralized compute-sharing platform*

cemlus/gridNode

Tools Used: Node.js, Redis, BullMQ, Socket.IO, PostgreSQL, Docker, gVisor, Terraform, AWS, Prometheus

- Built a decentralized platform distributing ML and video-rendering jobs across volunteer nodes: **Best Fit** bin-packing for placement, Trust Score reputation tie-breaking, and **heartbeat-based failure detection** with automatic job reassignment on node loss, over Redis/BullMQ queuing and S3 artifact storage.
- Executed untrusted third-party workloads in **gVisor-sandboxed containers** with live status streaming over Socket.IO – taken from empty repo to deployed in **two weeks**.
- Owned the deployment layer as well: **Terraform**-provisioned AWS (VPC, EC2, RDS, S3, least-privilege IAM), a **GitHub Actions** pipeline authenticating via OIDC with no static credentials, and **Prometheus/Grafana** instrumentation alerting on event-loop lag and connection-pool saturation.

Technologies

Languages: TypeScript, JavaScript, Python, C++, SQL, Bash

Backend & Data: Node.js, Express.js, PostgreSQL, MongoDB, Redis, BullMQ, Prisma ORM, Supabase, WebSockets, REST APIs

Infrastructure: AWS (EC2, RDS, S3, IAM), Terraform, Docker, GitHub Actions, Prometheus, Grafana, k6, Git, Linux